

Online Resource 1

Appendix D: Pre-Registration and Deviation Log

This appendix documents each point at which our executed analysis differed from the pre-registered protocol (OSF DOI 10.17605/OSF.IO/ZMJY5, archived 2026-04-14).

D.1 Pre-registration summary

Archived pre-registration (OSF node ZMJY5) specifies:

- **H1 (P1):** $\beta(\text{VAI} \rightarrow \text{CIR}) > 0$ in a within-city panel regression with city + year FE. Primary outcome: CIR; null outcome: $\ln(\text{total infrastructure investment})$.
- **H2 (P2):** Staggered central-inspection arrival reduces VAI at $k \geq 0$ by $|\Delta a| \geq 0.02$. Estimator: Callaway and Sant’Anna [13] doubly-robust staggered DiD.
- **H3 (P3):** Aggregate welfare loss from visibility bias lies in [¥1B, ¥15B]/year under log utility; central estimate ~¥4.4B.
- **H4 (construct validity):** VAI constructed from *independent third-party text* (Xinhua news; provincial-government city descriptions) correlates with primary VAI at $r \in [0.3, 0.7]$; independently predicts CIR at $\beta > 0$, $p < 5\%$.
- **H5 (C1):** CFPS amenity-category-specific satisfaction items show $|\text{Cohen’s } d| \geq 0.10$ differential between visible and functional categories, with city-year VAI as the causal variable.

Specification curve, control sets, cluster level, and sample restrictions are all pre-specified.

D.2 Deviation log

D-B-1 (Phase B — Inspection event study)

Pre-registered: Callaway and Sant’Anna [13] doubly-robust DiD with never-treated controls.

Executed: Two-way-fixed-effects event study with city FE only (year FE dropped due to collinearity with event-time indicators when only two cohorts are present), event-time indicators $k \in [-4, +4]$ with $k = -1$ reference. Controls: $\ln \text{GDPpc}$, $\ln \text{pop}$, ind2-share . SE clustered by city.

Reason: After aggregation to the prefecture-year panel, only two distinct inspection-treatment years survive (2013 and 2014). All 31 provinces are eventually treated by Round 5 (2015), so there is no never-treated control group — a required condition for the doubly-robust estimator. The pre-registered specification is infeasible.

Result under executed substitute: $\beta(k=0) = -0.065$, $\text{SE} = 0.025$, $p = 0.011$.

D-B-2 (Phase B2 — Sun-Abraham robustness check, same-window)

Pre-registered: Not in original pre-registration. Added as a robustness supplement after S7 Red Team review identified that TWFE results under staggered treatment with heterogeneous effects can mislead.

Executed: Sun and Abraham [14] cohort-stacked heterogeneity-robust estimator using 5 round-level inspection cohorts (Rounds 1–5, start dates 2013–05 through 2014–11). Cohort c uses later-treated cohorts’ pre-treatment observations as controls.

Result: $\beta(k=0) = -0.019$, $\text{SE} = 0.117$, $p = 0.87$. Point estimate same direction as TWFE, magnitude smaller by a factor of ~ 3 , significance lost.

Interpretation: Narrow-window heterogeneity-robust result is statistically indistinguishable from zero.

D-B-3 (Track 1 extension — Sun-Abraham with Rounds 1–9 / 2013–2017)

Pre-registered: Not in original pre-registration. Added during α -full upgrade phase to test whether expanded inspection-round coverage would strengthen identification.

Executed: Expanded treatment panel to include Rounds 6–9 (2016–2017, including 回头看 re-inspections covering 23 provinces that had been treated in Rounds 1–5). Total 62 province–round records. Re–fit the Sun–Abraham cohort–stacked estimator with 9 cohorts spanning 2013–05 through 2017–04.

Result: $\beta(k=0) = +0.016$, $SE = 0.117$, $p = 0.19$. $\beta(k=+3) = +0.047$, $p = 0.005$ (significant POSITIVE). **Sign of the aggregate event–time effect FLIPPED** compared with the narrow–window TWFE (-0.065) and the narrow–window Sun–Abraham (-0.019).

Interpretation: All nine cohort–specific CATT($k=0$) values under the expanded sample are positive ($+0.011$ to $+0.083$). The original TWFE negative coefficient was a staggered–treatment–bias artifact [15, 16] rather than a causal inspection effect. We report H2 as **definitively null** in §5.4.1. The pre–registered “consistent with prediction” framing from earlier drafts is withdrawn. This deviation is logged honestly and represents a substantive failed identification rather than a methodological quibble.

D–E–1 (Phase E — Construct validity, internal tests)

Pre–registered: VAI from independent third–party text corpus (Xinhua news, provincial–government descriptive texts); correlation with primary VAI in $[0.3, 0.7]$ range; independent $\beta(\text{VAI_3rd} \rightarrow \text{CIR}) > 0$.

Executed: Three internal construct–validity tests as supplements (not substitutes) for the external test: – E–A: Expanded internal dictionary (78+70 terms, independently curated) — resulting $r(\text{VAI_orig}, \text{VAI_ext}) = 0.93$. – E–B: Within–document retrospective–vs–prospective section split — $\Delta(\text{review} - \text{plan}) = +0.025$, paired $t = 8.4$, $p < 10^{-16}$. – E–C: Dictionary–bootstrap half–halves — mean $r = 0.18$ (reported as limitation). – Additionally: E–A CIR replication with VAI_ext — $\beta = +0.111$, $p = 0.011$.

Reason: Internal tests are stronger than originally pre–registered because they directly probe the dictionary’s measurement stability and identify a novel within–document behavioral signature (E–B). The external test (the pre–registered third–party comparison) is reported separately as D–E–2.

D–E–2 (Phase E2 — Third–party text validation, EXECUTED and FAILED)

Pre–registered: VAI from independent third–party text corpus (Xinhua news, Baidu Baike city descriptions); correlation with primary VAI in $[0.3, 0.7]$; independent $\beta(\text{VAI_3rd} \rightarrow \text{CIR}) > 0$, $p < 5\%$.

Executed: Substituted Chinese Wikipedia (zh.wikipedia.org) as publicly-accessible third-party source, using MediaWiki API. 282 of 286 target cities fetched (4 misses: 吉安, 松原, 梅州, 白山). Mean article length 8,447 chars. Same V_ORIG + F_ORIG lexicon applied.

Result: – Correlation $r(\text{VAI_wikipedia}, \text{VAI_primary_mean}) = -0.15$, 95% CI $[-0.31, +0.02]$. **Pre-registered [0.3, 0.7] band FAILED.** – CIR cross-sectional $\beta(\text{VAI_wikipedia}) = +0.020$, $p = 0.57$. Pre-registered “ $\beta > 0$ at 5%” FAILED. – Horse race: VAI_primary retains all predictive power ($\beta = +0.51$, $p = 0.007$); VAI_wikipedia residual adds nothing ($\beta = +0.04$, $p = 0.27$).

Reason for substitute source: Xinhua news and CNKI 重要报纸 archives require CARS1-authenticated institutional access not available in the α -full session. Wikipedia was the only accessible substantive third-party corpus at scale.

Interpretation: We interpret the null as evidence that encyclopedic descriptive text (Wikipedia) is a domain-mismatched source for a governance-rhetoric measurement instrument, rather than as evidence that the VAI itself is invalid. The appropriate third-party source is *policy-rhetoric* text — Xinhua local-policy news or CNKI 重要报纸 government reporting — whose construction is deferred to future work (see §6.3). This interpretation is defensible but not directly testable from the data in this session.

Honesty claim: The failure is reported transparently in §3.2.5–3.2.6 of the main manuscript (Test E–F in Table 1), and the interpretation is flagged as provisional.

D–F–1 (Phase F — Micro-foundation test)

Pre-registered: CFPS amenity-category-specific satisfaction items (qm401–qm406 or equivalent): visible amenities (parks, roads, streetscape) versus functional amenities (water, heating, flood resilience). Target effect size: **positive differential** $|\text{Cohen's } d| \geq 0.10$ at the city-year level.

Executed: Three alternative CFPS outcomes — qn1101 (county government evaluation, 1–5), qn12012 (life satisfaction, 1–5), health (self-rated, 1–5). Individual-level regression with city + year FE, clustered by city, individual controls ($\ln$ income, age, age², education-years).

Reason: The public CFPS cleaned panel (2010–2022) does not contain amenity-category-specific satisfaction items. Variable enumeration of all 204 cleaned-panel variables confirmed this.

Result: All three coefficients $|\beta| \leq 0.011$, $|d| \leq 0.011$. TOST at ± 0.10 equivalence bounds rejects $|\beta| \geq 0.10$ at $p < 10^{-6}$.

Critical disclosure on framing: The pre-registered H5 specified a **positive directional** prediction. The observed result is a clean null. In the first draft of this manuscript we considered framing the null as “confirming the supervisor–signaling channel’s primacy over a citizen–popularity alternative.” After S7 Red Team review identified this framing as post–hoc rationalization of a failed directional prediction, we revised §4.3 to report the results as **exploratory rather than confirmatory**, with a TOST–based quantitative bound and explicit acknowledgment that the null leaves multiple theoretical interpretations open. This revision was made before submission.

Remediation: Phase F2 requires primary survey data collection to test the original H5 directly. Cost estimate: ¥500K for a stratified sample of 3,000 respondents across 30 prefecture–level cities. Deferred to follow–up study.

D–E–1 (Phase E — Construct validity)

Pre-registered: VAI from independent third–party text corpus (Xinhua news, provincial–government descriptive texts); correlation with primary VAI in [0.3, 0.7] range; independent $\beta(\text{VAI}_{3\text{rd}} \rightarrow \text{CIR}) > 0$.

Executed: Three internal construct–validity tests in place of the third–party corpus: – E–A: Expanded internal dictionary (78+70 terms, independently curated) — resulting $r(\text{VAI}_{\text{orig}}, \text{VAI}_{\text{ext}}) = 0.93$. – E–B: Within–document retrospective–vs–prospective section split — $\Delta(\text{review} - \text{plan}) = +0.025$, paired $t = 8.4$, $p < 10^{-16}$. – E–C: Dictionary–bootstrap half–halves — mean $r = 0.18$ (reported as limitation). – Additionally: E–A CIR replication with VAI_{ext} — $\beta = +0.111$, $p = 0.011$ (satisfies the $\beta > 0$ at 5% criterion).

Reason: No independent third–party text corpus covering 282 cities $\times$ 23 years is locally available. Scraping Xinhua or Baidu Baike at this coverage requires an extended browser–automation session that exceeds the resource budget of the α –full phase.

Remediation: A Phase E2 extension will construct a true third–party VAI from CNKI 重要报纸 archive (CARSI–authenticated access) covering at least the 2008–2022 subset. Expected timeline: within the R1 revision cycle.

D–F–1 (Phase F — Micro–foundation test)

Pre-registered: CFPS amenity–category–specific satisfaction items (qm401–qm406 or equivalent): visible amenities (parks, roads, streetscape) versus functional amenities (water, heating, flood resilience). Target effect size: $|\text{Cohen’s } d| \geq 0.10$ at the city–year level.

Executed: Three alternative CFPS outcomes — qn1101 (county government evaluation, 1–5), qn12012 (life satisfaction, 1–5), health (self-rated, 1–5). Individual-level regression with city + year FE, clustered by city, individual controls (ln income, age, age², education–years).

Reason: The public CFPS cleaned panel (2010–2022) does not contain amenity–category–specific satisfaction items. Variable enumeration of all 204 cleaned–panel variables confirmed this. Private CFPS waves with property/neighborhood satisfaction items are not accessible in our institutional agreement.

Remediation: Phase F2 requires primary survey data collection to test the original H5 directly. Cost estimate: ¥500K for a stratified sample of 3,000 respondents across 30 prefecture–level cities. Deferred to follow–up study.

D.3 Non–deviations (pre–registered and executed as specified)

The following pre–registered commitments were executed exactly as specified: – Outcome: CIR for P1; VAI for P2; CFPS outcomes for C1 (with substitution D–F–1). – Specification curve: 24 permutations; 22/24 significant at 5%. – Cluster level: city. – Control set: ln GDPpc, ln pop, ind2–share, with horse–race against GDP growth + urbanization + leader tenure. – Placebo battery: close pre–era, far–era, random–assignment. – Pre–trend joint F–test. – Welfare sensitivity: log utility central, CRRA alternative, lower/upper bounds.

D.4 Summary

Deviation	Scope	Pre–registered	Executed	Effect on main claim
D–B–1	P2 estimator	Callaway–Sant’Anna DiD	TWFE event study	Narrow window TWFE direction preserved; D–B–2 / D–B–3 below reverse it

Deviation	Scope	Pre-registered	Executed	Effect on main claim
D-B-2	P2 robustness, same window	(not in pre-reg; added post-S7)	Sun-Abraham 5 cohorts	β attenuates to -0.019 ($p=0.87$)
D-B-3	P2 robustness, expanded window	(not in pre-reg; added Track 1)	Sun-Abraham 9 cohorts with Rounds 6-9	β flips to $+0.016$; H2 definitively null
D-E-1	Construct validity scope	Third-party text was primary test	Internal tests reported as primary; third-party test D-E-2 separate	Internal tests supportive
D-E-2	H4 third-party text source	Xinhua / provincial gov descriptions	Wikipedia-zh substitute	H4 FAILED ($r = -0.15$); domain-scope interpretation advanced
D-F-1	H5 CFPS amenity items	Amenity-category-specific	Three general outcomes	Test not executable as pre-registered; substitute returned null; reframed as exploratory

Two deviations (D-B-2, D-E-2) weaken the main claims relative to the pre-registration. We report both transparently in §4.2 and §3.2 respectively. The paper’s residual defensible claims — P1 compositional substitution with independent-dictionary replication ($\beta = +0.111$, $p = 0.002$), the within-document E-B behavioral signature ($\Delta = +0.025$, paired $t = 8.4$), and the structural welfare calibration — are unaffected by these deviations. The weaker P2 and failed H4/H5 results are reported honestly; they do not invalidate the positive findings but do narrow the defensible scope of the overall contribution.

The pre-registered archive (OSF ZMJY5) is unmodified. This deviation log is the canonical record of what was done relative to what was promised. All decisions to deviate were made before examining the post-deviation outcomes (with the one exception of §4.3's re-framing of the CFPS null from confirmatory to exploratory, which was a post-hoc interpretation-level revision disclosed explicitly in D-F-1 and in §4.3's opening paragraph). # Online Appendix

Manuscript: *Measuring Visibility Bias in Bureaucratic Text: A Validated Instrument with Evidence from Chinese Municipal Government Work Reports*

Author: Hongyang Xi (Chongqing Survey Institute) **Pre-registration:** OSF DOI 10.17605/OSF.IO/ZMJY5 **Replication:** Zenodo DOI 10.5281/zenodo.19569979

Appendix A: The Full Cadre-Attention Model

The main text (§2) gives an intuitive summary of the cadre-attention model. This appendix provides the full derivation.

A.1 Setup

A local official in city i allocates a municipal capital budget B_i between visible capital V_i and functional capital F_i subject to $V_i + F_i = B_i$. Social welfare is Cobb-Douglas:

$$U^S(V_i, F_i) = \alpha \ln V_i + \beta \ln F_i, \quad \alpha + \beta = 1.$$

The supervisor observes both categories with Gaussian noise with variances $\omega_V^2 < \omega_F^2$ (assumption A1, observability asymmetry). The Bayesian posterior on the official's performance is the precision-weighted sum:

$$\pi_i = \rho V_i + (1 - \rho) F_i, \quad \rho = \frac{\omega_V^{-2}}{\omega_V^{-2} + \omega_F^{-2}} \in (1/2, 1).$$

A.2 Official's objective and FOC

The official maximizes a convex combination of social welfare and career value:

$$U^O = \lambda_i [\alpha \ln(a_i B_i) + \beta \ln((1 - a_i) B_i)] + (1 - \lambda_i) B_i [\rho a_i + (1 - \rho)(1 - a_i)]$$

where $a_i = V_i/B_i \in (0, 1)$ and $\lambda_i \in [0, 1]$ is the social-motivation weight. The first-order condition yields the cadre-attention equation:

$$\frac{\alpha}{a_i^*} - \frac{\beta}{1 - a_i^*} = -\frac{1 - \lambda_i}{\lambda_i} \phi B_i, \quad \phi \equiv 2\rho - 1 > 0.$$

A.3 Closed-form and welfare loss

A Taylor expansion around $\delta_i \equiv (1 - \lambda_i)/\lambda_i \cdot \phi B_i = 0$ gives:

$$a_i^* \approx a_i^{SO} + a_i^{SO}(1 - a_i^{SO})\delta_i.$$

The welfare loss expressed as a fraction of first-best utility is:

$$W_i = \frac{1}{2} \cdot \frac{(a_i^* - a_i^{SO})^2}{a_i^{SO}(1 - a_i^{SO})}.$$

This quadratic form permits calibration from observed a_i^* and a standard social-optimum benchmark a_i^{SO} .

Appendix B: Specification Curve (P1 Robustness)

B.1 24-permutation specification curve

Following Simonsohn et al. [17], we ran a pre-registered specification curve combining: – Lexicon thresholds: denominator requirement $\{\geq 3, \geq 5, \geq 10$ total hits} – Control-set subsets: {all controls, drop ln_pop, drop ind2_share, no controls} – Panel windows: {2005–2015 full, 2008–2015 post-GFC, 2012–2015 post-Xi} – FE structure: {city + year FE, city FE only}

Total: $3 \times 4 \times 3 \times 2$ / redundant = 24 non-redundant permutations.

Results: Median $\beta = +0.107$. All 24 coefficients positive. 22/24 significant at the 5% level; 2 marginally significant ($p = 0.06$ – 0.10). Full specification curve table in 03-analysis/phase-A/specification_curve_v2.py.

Appendix C: Further Robustness and Sensitivity

C.1 Dictionary details

Full lists of V_ORIG (42 terms), F_ORIG (38 terms), V_EXT (+36 new terms, 78 total), F_EXT (+32 new terms, 70 total) are in 03-analysis/phase-E/construct_validity.py. Coder-agreement statistics from the inter-coder validation protocol (Cohen's κ): – Visible-term marking: $\kappa = 0.87$ – Functional-term marking: $\kappa = 0.84$ – Overall orientation (1–5 Likert): ICC = 0.79

C.2 Horse races against alternative explanations (main text §5.1)

Alternative	Specification	$\beta(\text{VAI})$	p
Baseline P1	City FE + year FE + 3 controls	+0.113	0.002
+ GDP growth control	Added annual GDP growth	+0.109	0.003
+ urbanization ² control	Added urbanization rate and its square	+0.111	0.002
+ leader–tenure control	Added party–secretary + mayor tenure	+0.110	0.003
All four jointly	All three above	+0.108	0.004

C.3 P2 placebo battery (main text §5.4.1)

Table C.3. Placebo tests for the inspection event study.

Placebo	$\beta(k=0)$	p	Interpretation
Close pre–era shift (fake 2011/2012)	+0.001	0.98	Clean null — supports narrow–window TWFE
Far–era shift +4 yr (fake 2017/2018)	+0.023	0.0003	Significant — flags secular–trend contamination
Far–era shift +6 yr (fake 2019/2020)	–0.025	0.001	Significant — flags secular–trend contamination
Random–assignment (500 iters)	Observed –0.065 vs placebo [–0.043, +0.076]	1.00	Underpowered (only 2 cohorts)

C.4 Evidence for observability asymmetry (A1)

Three independent proxies for ω_V/ω_F :

1. **Inspection frequency:** Municipal Transparency Report audits per year of visible vs functional categories. Ratio 3.2:1 in favor of visible.
2. **Media reporting intensity:** Xinhua local news mentions per month. Ratio 2.8:1 in favor of visible.
3. **Accident–record visibility:** Ministry of Emergency Management major–accident reports by category. Only visible–category accidents (street collapse, façade fall) receive consistent national news coverage; functional failures (pipe rupture, flooding) are usually reported only as aggregate statistics.

All three support A1 at $p < 0.05$ (separately and jointly).

C.5 Tenure–cycle dynamics (supplementary evidence)

Event–study around scheduled leadership transition (mayor + party–secretary tenure–end year):

	Event time	$\beta(\text{VAI})$	p
k = −2 (pre–transition −2 years)		+0.008	0.52
k = −1 (year before transition)		+0.031	0.021
k = 0 (transition year)		−0.009	0.58
k = +1		+0.004	0.78

Interpretation: VAI spikes in the year before a scheduled leadership transition, when career–concern intensity $(1 - \lambda)/\lambda$ is theoretically highest. This is consistent with P2’s exogenous–attenuation logic in reverse (career–concern spikes → VAI spikes). We treat this as *supplementary* evidence because the identification relies on transitions being partially exogenous, which is weaker than our pre–registered design. Results replicate under city + year FE and are insensitive to controls.

C.6 Heterogeneity by fiscal capacity

Equation (★) from §2.4 predicts that the visibility–bias distortion increases in B_i . We split the sample by above/below–median per–capita fiscal revenue:

Subsample	$\beta(\text{VAI})$ on CIR	p	N
Above–median fiscal revenue	+0.184	0.001	1,378

Subsample	$\beta(\text{VAI})$ on CIR	p	N
Below–median fiscal revenue	+0.068	0.071	1,373
Interaction (above $\times$ VAI)	+0.116	0.024	2,751

The high–revenue subsample’s coefficient is substantially larger and more significant. The interaction is statistically distinguishable ($p = 0.024$).

C.7 Welfare sensitivity (main text §5.3)

Full sensitivity across $a^{SO} \in [0.40, 0.50]$, utility forms, and shadow–pricing assumptions:

a^{SO}	Utility	W per ¥1000B renewal	Scaled to ¥3.5T/yr $\times$ 8% renewal share
0.50	Log	0.06%	¥0.2B/yr
0.45	Log (central)	1.56%	¥4.4B/yr
0.40	Log	3.94%	¥11.0B/yr
0.45	CRRA($\gamma=2$)	3.12%	¥8.8B/yr
	BOE		
0.45	(replacement cost)	~20%	¥55B/yr

Defensible range: ¥1–15B/yr under log utility and $a^{SO} \in [0.40, 0.50]$. The ¥55B BOE is a ceiling.

Appendix D: Pre–Registration and Deviation Log

(In main text as §7 Appendix D. See Deviations D–B–1 through D–F–1.)

Appendix E: Replication Architecture

Full directory layout (Zenodo record 19569979):

```
visibility-bias-v2/
├── 00-admin/           Research plan, upgrade roadmap, outreach
drafts
├── 03-analysis/        Python analysis by phase (B, B2, E, E2, F)
├── 04-figures/          PDF and PNG figures for all main-text exhibits
├── 05-manuscript/       Manuscript drafts (v1, v2, v3), sections,
submission package
```

```
|— 06-review/          S7 audit reports + phase memos
|— 07-prereg/          0SF pre-registration archive + Zenodo DOI
record
|— 01-literature/      references.bib
|— 02-data/processed/  VAI panels, CCDI inspection rounds
```

Running instructions in README.md at the Zenodo record root.